



COMP333: Date Base Systems

Phase #2

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Project Title: LTC E-Commerce Website

Client: Light for Technology & Computer (LTC) – Ramallah

Project Scope and Description

Light for Technology and Computer (LTC) is a computer and networking store located in Ramallah that sells laptops, PC components, accessories, and networking devices. The goal of this project is to develop an e-commerce website that allows customers to browse products online, add items to a shopping cart, and place orders securely.

Administrators will have access to a management dashboard that allows product updates, tracking inventory, managing orders, and viewing customer details.

The system will include:

- Customer registration and login
- Product browsing and searching
- Categories and filtering
- Shopping cart and checkout
- Order and delivery tracking
- Admin panel for product and order management

The database will ensure efficient storing and retrieving of records related to customers, products, orders, suppliers, and staff.

Client Information

Client Name: Light for Technology & Computer (LTC)

Location: Ramallah, Palestine

Industry: Technology, Computers & Networking Equipment

Client Needs: Expand business to an online environment, automate product and order handling, secure user data, and improve accessibility and sales.

Technology Stack:

The application will be developed on the Windows OS. The frontend user interface (UI) will be designed using HTML for structure and CSS for styling. For backend functionality, the Flask framework in Python will handle server-side logic, while MySQL will serve as the relational database for data storage and management.

Expected Entities and Relationships:

Entities	Attributes
Branch	<u>Branch ID</u> , location, phone number
Employee	<u>Employee ID</u> , Branch ID, name, email, date of birth, salary, phone number
Customer	<u>Customer ID</u> , name, phone number, date of birth, Address
Product	<u>Product ID</u> , Name, StockQuantity, price(), Description
Warehouse	<u>Warehouse ID</u> , Branch ID, Product ID, Quantity
Order	<u>Order ID</u> , Customer ID, Branch ID, Product ID, Delivery Address, order date, total Amount(), status
sale	<u>Sale ID</u> , sale percentage, duration, Product ID
Payment	<u>Payment ID</u> , method name, order ID
User	<u>User ID</u> , password

Relations:

- Customers can place orders that include multiple products and choose their preferred payment method.
- Customers can create accounts to log in, and employees also have login accounts to track actions such as managing products or processing orders.
- Our store has multiple Branches.
- Each Branch is managed by exactly one Employee.
- Every Employee works in exactly one Branch, and each Branch has many Employees.
- A Manager is just an Employee who manages a Branch (not every Employee is a manager).
- A warehouse supplies many branches, and Each Branch has one warehouse
- Each Warehouse stores many Products in its stock, and a Product can be available in many Branches.
- All sellable items live in Products(ProductID, Name, ...).
- Products have Categories: Laptops, PCs, MobilePhones, Watches, Accessories, HeadPhones, NetworkingDevices, StorageElements, Printers.
- Sales records apply to multiple Products; a Product can have one Sale.
- A Customer can place many Orders; each Order belongs to one Customer.
- Each Order is handled by one Branch (BranchID in Orders), and a Branch can handle many Orders.
- An Order can include multiple Products, and a Product can appear in many Orders.

Queries:

CREATE (Adding new data)

1. Add a new product to the system.
2. Add a new customer account.
3. Add a new order made by a customer.
4. Add new stock quantity for a product in a branch.

READ (Retrieving and viewing data)

5. Total number of branches.
6. Manager information for each branch.
7. Number of employees in each branch and overall total.
8. Products available in a specific branch.
9. Quantity of a specific product in a specific branch and across all branches.
10. Total number of orders made by a specific customer.
11. Total revenue from orders per month or year.
12. Best-selling product in each branch or overall.
13. All products purchased by one customer last month.
14. Out-of-stock products in a specific branch.
15. Total number of users registered on the website.
16. Total number of laptops in stock in a specific branch.
17. Available products in all branches.

UPDATE (Modifying data)

18. Update the price of a product.
19. Update product stock after a new shipment.
20. Update a customer's contact information.
21. Update an order status such as "Pending" to "Shipped" or "Delivered."

DELETE (Removing data)

22. Delete a discontinued product.
23. Delete a canceled order.
24. Delete a customer account upon request.

ER Diagram

